

Residual Debiasing for Robust Natural Language Inference

Abstract

In this project I investigated the extent to which SNLI models rely on hypothesis level artifacts instead of (premise, hypothesis) reasoning. To isolate these effects, I compared a standard NLI classifier against a hypothesis only model that receives no premise information. The hypothesis only model reaches a 68.84% accuracy suggesting that the hypothesis itself contains strong label predictive patterns. The full model reached 88.28%. The large gap between the full model and the hypothesis only baseline suggests that most of the performance comes from exploiting statistical regularities rather than its true inferential capability. To lessen the gap, I used a residual debiasing approach for ELECTRA-small. My results show that this debiased model preserves overall accuracy while reducing dependence on hypothesis only cues.

1 Introduction

Natural Language Inference (NLI) benchmarks such as the Stanford NLI (SNLI) dataset are designed to test a model’s ability to determine whether a hypothesis is entailed, contradicted, or unrelated to a natural language premise (Bowman et al., 2015). Transformer based models achieve strong performance on these types of tasks, but studies show that higher accuracy often shows the exploitation of annotation artifacts rather than the reasoning (Gururangan et al., 2018; Poliak et al., 2018).

In this project, I empirically examined this issue by comparing a standard NLI model with a hypothesis only classifier. I measured how accurately labels can be predicted without access to the premise, following prior work on hypothesis only baselines (Poliak et al., 2018).

A large gap between the full model and hypothesis only model indicates the degree to which dataset artifacts inflate reported accuracy. After I quantified the bias I implemented residual debiasing. It

decomposes predictions into a bias term and a reasoning term (He et al., 2019). The bias model is frozen and the full model is trained to predict the residual signal that remains after subtracting bias model logits. This prevents the full model from relying on simple hypothesis only shortcuts.

My goals were to provide an analysis that shows that SNLI contains strong labeling artifacts, full implementation of residual debiasing for ELECTRA-small (Clark et al., 2020), and empirical results that show that debiasing maintains accuracy while reducing reliance on hypothesis only cues.

2 Approach

2.1 Dataset and Preprocessing

I used the SNLI dataset with invalid labels removed, there were around 549,000 examples (Bowman et al., 2015). The validation set contains around 10,000 examples. For the baseline model, training follows standard ELECTRA tokenization using paired (premise, hypothesis) inputs (Clark et al., 2020). For the hypothesis only model, the premise is replaced by an empty string while retaining full tokenization of the hypothesis, as it is in prior work on hypothesis only baselines (Poliak et al., 2018). Both models use max-length 128 padding and truncation.

2.2 Baseline Model

First I fine-tuned an ELECTRA-small discriminator model on the SNLI dataset for one epoch following the starter code provided in the repository. I chose to do one epoch because doing three seemed unnecessary. I deemed it unnecessary because I did try three epochs at first, but the logs showed that at each interval there was minimal change through the first epoch, so for the sake of time and efficiency I stuck with one epoch. The model achieved an accuracy of 88.28% on the validation split. This suggests that the model performs well on the bench-

mark, but studies show that such high performance could be due to dataset artifacts rather than true reasoning capability (Poliak et al., 2018; Gururangan et al., 2018). To investigate this I conducted a hypothesis only analysis and examined several classes of examples known to expose superficial heuristics, I go into further detail in section 3.

2.3 Hypothesis Only Baseline Model

To measure the degree of annotation artifacts in SNLI I trained a second model that receives only the hypothesis as input with the premise replaced by an empty string. This model did not have access to the needed information for true inference. Therefore, performance above chance (33%) is evidence of false correlations.

Even though it lacked the premise entirely, the hypothesis only model achieved 68.84% accuracy, which is nearly 20 points below the full model but more than double the random chance. This result strongly suggests that the hypothesis text alone contains label predictive features that the model can exploit. Because there is a large gap between the full model (88.28%) and the hypothesis only model (68.84%), we can assume that a substantial portion of the baseline’s performance comes from memorizing statistical regularities instead of performing actual entailment reasoning.

These findings align with research that showed SNLI contains numerous artifacts such as cue words like “nobody” and “never”, lexical patterns correlated with specific labels, and stereotyped contradiction templates, like polarity reversals (Gururangan et al., 2018). My results replicate this occurrence.

2.4 Method Overview

My analysis in Part 1 revealed that SNLI contains strong hypothesis only artifacts. The model trained without access to the premise and achieved 68.84% accuracy, which confirms that the hypothesis by itself contains substantial label predictive cues. To reduce the model’s reliance on these artifacts I applied the residual debiasing method (He et al., 2019).

Residual debiasing separates two sources of model behavior. The bias component, which is captured by a frozen hypothesis only model:

$$f_b(h) \quad (1)$$

And the intended reasoning component, which is the part of the prediction not explained by the bias

model. I trained a new full model:

$$f_m(p, h) \quad (2)$$

whose outputs are combined with the bias logits to form the residual representation:

$$\text{logits}_{\text{debias}} = f_m(p, h) - f_b(h) \quad (3)$$

The training objective was then applied to these debiased logits, which forced the main model to explain whatever portion of the label distribution the hypothesis only model could not explain. This encouraged the model to make use of the premise and it also reduced reliance on annotation artifacts. The bias model:

$$f_b \quad (4)$$

is frozen throughout training so that the debiased model doesn’t relearn the same shortcuts.

2.5 Residual Debiasing Implementation

I implemented residual debiasing using a modified forward pass in a custom ElectraForSequenceClassification wrapper. During every training step the main ELECTRA model encoded the full input pair, premise and hypothesis. The frozen hypothesis only model encoded the hypothesis alone with an empty premise token. The logits from the main model were subtracted from the logits of the hypothesis only model, and cross entropy loss was computed on the residual logits only.

I trained it for one epoch to match our baseline setup, using the same batch size, tokenization, and preprocessing pipeline. Both training and evaluation were run in Google Colab with T4 GPU.

3 Error and Heuristic Analysis

I examined specific classes of hypotheses to understand how to characterize the model’s artifacts.

3.1 Negation Heuristic

Hypotheses containing negation tokens disproportionately receive the contradiction label in SNLI. Negation tokens are for example: “no,” “not,” “never,” “nobody,” and “nothing.” Models often learn a simplistic rule, which is if a hypothesis contains negation then predict contradiction. This explains why hypothesis only models achieve high performance, negation words show up far more often in contradiction examples. Manually inspecting failure cases showed that the full model predicted contradiction even when the negation did not contradict the premise, for example “A man is running” in contrast to “A man is not unhappy”.

3.2 Lexical Overlap Heuristic

High lexical overlap between premise and hypothesis typically pushes the model toward predicting entailment. I found that many full model errors fall into cases where nearly all content words overlap, the label is contradiction or neutral, and the model incorrectly predicts entailment. This suggests reliance on surface lexical similarity rather than semantic reasoning.

3.3 Antonym and Polarity Heuristic

For pairs containing antonyms like “hot” and “cold” or “standing” and “sitting”, the model performed well when the antonyms appeared explicitly. The model struggled when the contradiction required multi-step reasoning like “The man is awake” and “The man is not sleeping.” The hypothesis only model often predicted contradiction for these cases based solely on polarity cues.

3.4 Length and Template Artifacts

Contradiction hypotheses in SNLI tend to be shorter and follow consistent syntactic templates. Entailment hypotheses often repeat premise structures with minor lexical changes. The model appears to exploit these patterns.

3.5 Contrastive Example Failures

I constructed minimal contrast examples by altering a single word in the hypothesis. Many of these exposed weak decision boundaries. For example, the original was “A dog is running through a field.” The hypothesis was “A dog is running” (entailment). And the contradiction was “A dog is sleeping.” The full model correctly classifies the original but fails on several contradictions. This indicates reliance on memorized lexical associations rather than semantic understanding.

4 Results

4.1 Accuracy Across Models

Model	Validation Accuracy
Baseline ELECTRA	0.8828
Hypothesis-Only Model	0.6884
Debiased Model	0.8803

Table 1: Validation accuracies for baseline, hypothesis-only, and debiased models.

Even though the hypothesis only logits were subtracted, the debiased model kept nearly the full accuracy of the baseline. This matches prior studies showing that residual methods tend to preserve in-domain performance while improving robustness (He et al., 2019).

4.2 Improvements

Although accuracy changes only by a little, the debiased model demonstrated improvements in artifact sensitive slices. In negation cases the baseline often defaults to predicting contradiction when “not”, “never”, or “nobody” appear in the hypothesis. The debiased model corrects several similar cases, which reduces the false positives. With lexical overlap contradictions, examples with high word overlap but contradictory meanings like “A man is sitting” and “A man is standing” are handled more reliably. With antonym only contradictions, cases where the contradiction is subtle but requires semantic reasoning improved, as it did with “awake” and “sleeping.” Regarding contrast sets, minimal word swaps, such as “running” and “sleeping” produced more consistent decision boundaries than the baseline. These improvements indicate that the model is less reliant on shallow heuristics and more tuned to assume hypothesis relationships.

The residual debias method unfortunately does not eliminate all artifacts, only those captured by the hypothesis only model. It is apparent that it is more difficult to disentangle biases that depend jointly on (premise, hypothesis) interactions. But aside from that this method is helpful and complements other strategies like contrast sets and adversarial testing (Ribeiro et al., 2020). It also results in giving us a classifier that uses the premise more consistently, relies less on vague hypothesis cues, and it maintains high in-domain accuracy (He et al., 2019).

4.3 Summary

To summarize my analysis and results, I found that SNLI contains strong hypothesis only artifacts, as seen by the 68.84% accuracy of the hypothesis only classifier. Additionally, the full model relies heavily on superficial heuristics, especially negation, lexical overlap, and polarity cues. Furthermore, the 19.44 point gap between the full and hypothesis only models indicates that almost a quarter of the prediction signal can be explained by hypothesis only artifacts alone. This confirms that the high SNLI accuracy of many NLI models partially

reflects dataset shortcut exploitation, which is in agreement with (Gururangan et al., 2018; Poliak et al., 2018). And finally, the contrast set evaluation revealed that the model’s performance degrades under minimal perturbations, which is consistent with prior findings on dataset artifacts (Ribeiro et al., 2020).

5 Conclusion

In conclusion, residual debiasing essentially mitigates hypothesis only shortcuts without significantly degrading overall in-domain performance. This method encourages the model to rely on the premise for decision making instead of memorizing statistical regularities in the hypothesis. While the accuracy gains are small, the decrease in false correlations is meaningful and supports the use of debiasing methods for NLI benchmarks.

References

- Samuel R. Bowman, Gabor Angeli, Christopher Potts, and Christopher D. Manning. 2015. A large annotated corpus for learning natural language inference. In *Proceedings of EMNLP 2015*.
- Suchin Gururangan, Swabha Swayamdipta, Omer Levy, Roy Schwartz, Samuel R. Bowman, and Noah A. Smith. 2018. Annotation artifacts in natural language inference data. In *Proceedings of NAACL 2018*.
- Adam Poliak, Jason Naradowsky, Aparajita Haldar, Rachel Rudinger, and Benjamin Van Durme. 2018. Hypothesis-only baselines in natural language inference. In *Proceedings of the 7th Joint Conference on Lexical and Computational Semantics (*SEM 2018)*.
- Kevin Clark, Minh-Thang Luong, Quoc V. Le, and Christopher D. Manning. 2020. ELECTRA: Pre-training text encoders as discriminators rather than generators. In *Proceedings of ICLR 2020*.
- He He, Sheng Zha, and Haohan Wang. 2019. Unlearn dataset bias with model decomposition. In *Proceedings of EMNLP 2019*.
- Marco Tulio Ribeiro, Tongshuang Wu, Carlos Guestrin, and Sameer Singh. 2020. Beyond accuracy: Behavioral testing of NLP models with CheckList. In *Proceedings of ACL 2020*.